Herbal Knowledge Intelligence System - AI Assistant for Sri Lankan Medicinal Plants

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
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ABSTRACT

Medicinal plants of Sri Lanka form a component of the rich flora of the country, which has contributed to the evolution of Ayurveda in Sri Lanka, a traditional system of medicine practiced in Sri Lanka for the last 3000 years. There are approximately 1400 species of medicinal plants reported in Sri Lanka and there is a decline in the indigenous knowledge of medicinal plants as the traditional practitioners who have years of experience have not been able to explain the concepts to the young practitioners mainly due to the age factor of having passed 55 years and above. Further, current ontologies are primarily query able via Simple Protocol and RDF Query Language (SPARQL) which requires expertise in computers. Also, current chatbots are based on pattern matching over very thin knowledge bases which is far from the in-depth domain knowledge required for this application.

We developed an Herbal Knowledge Intelligence System by integrating a Conversational Artificial Intelligence (AI) assistant called Herbal Knowledge Assistant (HKAS) based on Retrieval-Augmented Generation (RAG) framework that delivers correct and relevant information about medicinal plants available in Sri Lanka in both English and Sinhala languages. The database is a list of 20-30 plants from the medicinal flora of Sri Lanka which are commonly used by traditional local healers. This database is connected with large language models and the desired functionality is achieved using LangChain for pipeline orchestration, OpenAI's for multilingual semantic search, ChromaDB as a vector store and GPT-3.5-turbo for response generation. Users can ask questions about their required information by using free-form natural language questions. We will be using a Flask backend for a Flutter-based mobile frontend along with additional plant identification components.

Our project will yield a fully functional and content relevant and reliable prototype. This outcome will contribute to the realization of Sustainable Development Goals 3 - Good Health and Well-being, and 4 - Quality Education. Our goal is to secure the traditional knowledge of local medicine relevant to indigenous populations, and to develop an educational resource that is accessible to everyone. We anticipate revenue streams through freemium pricing, subscription to hospitals and schools, and aggregated data in anonymized form for research institutions.

Keywords: *Retrieval-Augmented Generation, Medicinal Plants, Ayurveda, Conversational AI, Sinhala Language Processing*

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LIST OF ABBREVIATIONS

Abbreviations	**Description**
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
LLM	Large Language Model
LM	Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation

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1 INTRODUCTION

1.1 Background and Context

Medicinal plants are abundant in our country, Sri Lanka. The Department of Ayurveda says there are about 1,400 medicinal plant species in our country [17]. They were traditionally used for curing many diseases in Ayurveda which has a long history of more than 3000 years in Sri Lanka. It is traditionally believed that each plant of medicinal origin possessed a special type of energy which, when ingested or applied to the required part of the human body, restored health and vitality. Medicinal plants like Kohomba (Neem), Komarika (Aloe vera), Aralu (Terminalia chebula), Bulu (Terminalia bellirica), Nelli (Phyllanthus emblica), Venivel (Coscinium fenestratum) are many examples of those valuable plants that were used in our Ayurveda medicine centuries and centuries ago [1], [19].

The very fact that this information exists today means it is at risk. The ethnic names of these plants, their various plant parts used for medicine and methods of preparation, as well as specific diseases for which they are employed, are almost invariably recorded as part of the indigenous knowledge that is in the form of oral tradition, or that has been documented by varying degrees in ancient literature [18]. Presently, less than 30% of the population can name some medicinal plants that grow in their gardens or backyards. Indigenous knowledge related to medicinal plants has been held mainly with practicing Ayurveda healers in the upper age group of 55 years and above. These sources are rapidly diminishing as most traditional practitioners in this age group are passing away due to modernization and modern health care, and passing on the traditional knowledge to the young is taking place very little [19]. Added to this is the fact that all these medicinal plants are being continuously uprooted and destroyed in huge quantities as ‘weeds’. The population at large has no appreciation of the great contribution these simple weeds play towards health and medicine.

Current information of medicinal plants is not readily available to the common man. There are only a handful of books which gives information of medicinal plants [18, [19]. These books are very costly, hard to get and again not very helpful as they cannot be searched. Again, current information is scattered all over the Internet in various web sites, forums and many research papers. Again, the reliability of these web sites may differ, and one may not get correct information of the plants they are searching for. Many web sites use English as their main language and as the knowledge of English is not high in our country the local population will

not be able to use them as much as they will be able to use a web site having knowledge in Sinhala.

This issue was also addressed by digital means. For instance, Jeyananthan et al. [1] used Protégé for creating an ontology for the Sri Lankan medicinal plants. It provides a knowledge base for plants, their parts, diseases, process of preparation of medicines and mixtures of them. But the user should know the SPARQL (Simple Protocol and RDF Query Language) queries in order to access this information. While the common man, as well as even many professionals cannot learn SPARQL queries, it also reduces the usage of such databases. Another approach was introduced by Jayalath et al. [2] in the form of a Sinhala chatbot with an Artificial Intelligence Markup Language (AIML)-based application for giving information on Ayurveda and medicinal plants. Their application was a pattern-matching-based model that is able to answer user input as long as it can match the already given patterns. Therefore, it is limited to the given inputs. More recently, Senevirathne et al. [3] also produced a mobile app that has a similar basic intent-based chatbot as well.

Currently available solutions suffer from one of two weaknesses. Ontologies have very deep knowledge, but this knowledge is inaccessible. Most modern chatbots have a easy interface to communicate with the user but the knowledge they contain is usually limited and based on pattern matching. Currently there is no solution that has the depth of knowledge sources and at the same time the ease and simplicity of natural language conversation interfaces.

We are in an exciting era for Artificial Intelligence with breakthroughs in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) in particular. RAG consists of two sub-modules: - a retrieval module that queries a large-scale knowledge base to find relevant passages; - a generation module (the LLM) that generates an answer based on the relevant passages it extracts from the knowledge base [4], [6], [7]. This approach has several benefits: The output is linked to factual information in a well-maintained database, which avoids “hallucinations” (LLM fabricates information that does not exist in the database) questions can be efficiently processed since the retrieval module can return relevant passages just-in-time [5]. RAG can be extended to support multi-languages with multilingual embedding models, which provides a scalable way for supporting diverse language outputs [6].

The Herbal Knowledge Intelligence System developed in this research study is a conversational AI assistant that provides information to the users on medicinal plants that are found in Sri Lanka in both English and Sinhala languages. To achieve this, it proposes the RAG approach

by combining a database of plant information curated by experts and advances in LLMs that enhance the natural language understanding and generation capabilities. The user can input the required information in a free form language as “What is Kohomba used for?” and will get the relevant answers.

As of now we are planning to use this system as a proof of concept with a starting list of around 20-30 of the most used medicinal plants found in Sri Lanka. We will use the following python modules: - LangChain (RAG) - Multilingual sentence transformers to index our data to perform searches - ChromaDB (vector store) – OpenAI’s GPT-3.5-turbo for LLM. We will use a Flask backend with a Flutter frontend. This will link to the larger system of the project that would be later linked to our plant identification system.

The proposed project will utilize recent advancements in Artificial Intelligence (AI) and blend them with principles of ancient medicine to safeguard the wealth of traditional knowledge in Ayurvedic drugs and to provide a contemporary information service which can be accessed by future users of Ayurveda in their local language.

This project aligns with Sustainable Development Goal (SDG) 3: Good Health and Well-being by enhancing access to traditional medicine knowledge and SDG 4: Quality Education by serving as a learning resource for students and the public.

1.2 Literature Review

Several researchers have explored ways to preserve and share herbal plant knowledge using technology.

Research 1: Ontology for Sri Lankan Medicinal Plants

Jeyananthan et al. [1] have built an ontology for medicinal plants found in Sri Lanka using the Protégé tool. The classes in the ontology consist of Plants, Plant Parts, Diseases, Preparation Methods, Mixtures and Application Categories with the corresponding relations. The system covered about 200 plants with their attributes. Even though the knowledge is represented in a very structured way using an ontology, the users have to know how to write Simple Protocol and RDF Query Language (SPARQL) queries, which is not familiar to all users. Therefore, the system is not user friendly.

Research 2: Ayurvedic Knowledge Sharing Platform with Sinhala Virtual Assistant

An experiment was carried out by Jayalath et al. [2] for automatic plant identification using several deep learning models such as Convolutional Neural Network (CNN) and to create a chatbot application in the language of Sinhala of Sri Lanka using AIML and pattern-matching. This application provides relevant information on Ayurveda and medicinal plants related to a given question by pattern-matching with user questions. They claim that "The system analyzes the question that user asks and will provide answers according to that" to the extent that the question matches the patterns they have defined. The limitation is that it can not cope with the questions that do not match with any of the patterns [2].

Research 3: Mobile-based Assistive Tool to Identify & Learn Medicinal Herbs

There was a Mobile App developed by Senevirathne et al. [3] which includes three sections: a plant identifier that utilised Fast Region CNN, a 3D viewer built with Unity, and a Sinhala chatbot that utilised AIML. The AIML Chatbot used intent classification to differentiate between different queries about the same item by using the variations in the question (for e.g. "what is komarika" vs. "tell me about komarika"). It is limited to the depth of the pre-defined information available in the database. [3]. An accuracy of 96% was reported but this was for very limited information and only up to the intent of the predefined items.

Research 4: Applications of LLMs in Healthcare and Medicine

In a study led by Abd-alrazaq et al. [5], the researchers investigated the possibilities and challenges of using LLMs in the context of medical education, while highlighting the potential of LLMs in providing domain-specific knowledge-based applications while addressing the issues of accuracy and hallucinations. Thirunavukarasu et al. [11] conducted a comprehensive review on the use of LLMs in medicine, providing a basic groundwork for the conversational AI in the healthcare sector. Tanasă et al. [12] proposed a multi-agentic AI-powered virtual assistant for medical clinics using LLM and RAG.

Research 5: RAG for Knowledge Management

Pondel et al. [6] discussed some approaches to build the knowledge base using LLM and RAG. Their discussion is very relevant to our work of knowledge management of medicinal plants. They said that an appropriate knowledge base is required to ensure high accuracy of the output

of the search model and also, they gave several methods to find the optimum chunk size and optimum embedding for the knowledge base.

Research 6: RAG and LLM Technologies for Domain-Specific Applications

Sundar et al. [7] evaluated the feasibility of using RAG and LLM models for the purpose of educational assessment. In another contribution, Shafi et al. [8] proposed a RAG-based multimodal approach for developing a behavioral therapist. In this contribution, D'Urso et al. [9] proposed new approaches to Language Model (LM) design for efficient intelligent system configuration. In this contribution, Chia et al. [10] proposed the use of Language and Multi-Task Learning Models for developing an Intelligent Virtual Assistant (IVA) for smart irrigation.

Research 7: Foundational NLP and LLM Technologies

The standard model for LLMs is the Transformer architecture proposed by Vaswani et al. [13]. The BERT model proposed by Devlin et al. [14] is one of the most successful models in the field of NLP. Brown et al. [15] in their paper show the few-shot learning capability of large language models with GPT-3. Finally, open-source LLM, Llama 2, proposed by Touvron et al. [16] in this paper provides a good trade-off between performance and the need for large resources, which is very relevant to our use case, hence being the model chosen for this project.

Table 1-1 Summary of Existing Approaches

Approach	**Strengths**	**Limitations**
Ontology [1]	Well-structured, comprehensive knowledge	Requires SPARQL queries – not user-friendly
Pattern-matching chatbot [2]	Simple to implement, Sinhala support	Cannot handle novel questions; limited to pre-defined patterns
Intent-based chatbot [3]	Understands query variations	Limited knowledge depth; no retrieval from comprehensive sources

1.3 Research Gap

Although numerous studies have been carried out to preserve and utilize knowledge on medicinal plants of Sri Lanka, there are many unexplored areas. This subsection briefly explains the research gaps which this research study aims to address.

Gap 1: No RAG Implementation for Sri Lankan Herbal Medicine

The Relational Abstract Generator (RAG) model has been primarily fine-tuned for applications in the domains of education [4], healthcare [8], [11], [12] and agriculture [10]. Additionally, it was used in the context of knowledge management in [6]. However, there is no prior work that utilized RAG for the purpose of maintaining medicinal plant knowledge specific to Sri Lanka. Hence, there is a huge scope of bridging authoritative literature on Sri Lankan medicinal plants and the work done by Jeyananthan et al. [1] with conversational AI by utilizing RAG.

Gap 2: Limited Semantic Understanding of Sinhala in Medical Domain

The existing Sinhala language-based medical plant search chatbots are based on pattern-matching (AIML) or simple intent classification [2], [3]. They do not have the ability to have semantic understanding of the input questions and therefore cannot serve to the needs of the users effectively. Questions like “How to use Komarika” and “How should I take Komarika” should be identified as “Usage method of the plant” though the words are different and chatbots have to have different patterns for different wordings currently. The intended work in this research will be based on modern multilingual embedding models [5], [6].

Gap 3: Static Knowledge vs. Dynamic Retrieval

Current approaches to represent knowledge about Sri Lankan medicinal plants are static and need to be manually updated, for example, the ontology proposed by Jeyananthan et al. [1] needs to be manually enriched with new medicinal plants and attributes. Although a very complete and well-explained approach, the need to manually introduce the new medicinal plants and their attributes is a time consuming and human dependent process. Also, each time a new question type needs to be addressed by a pattern-matching chatbot, new patterns for this specific type of question need to be manually added [2]. Contrarily, RAGs allow dynamic updates. Introducing new information to a vector database such as RAG, only needs introducing new token chunks to the database, without the need to retrain the embedding model, or manually generate and integrate new patterns.

Gap 4: No Evaluation Framework for Ayurvedic Conversational AI

There are no benchmark evaluations and metrics available for conversational AI in the Sri Lankan Ayurvedic domain. Thus, there is need to identify the metrics that considers technical metrics as well as expert validation and user experience for such a chatbot.

Table 1-2 Comparison of Existing Systems and Proposed System

Research Aspect	**Ontology [1]**	**Pattern Chatbot [2]**	**Intent Chatbot [3]**	**Proposed System**
Natural Language Interface	No	Yes	Yes	Yes
Handles Novel Questions	No	No	Limited	Yes
Knowledge Depth	High	Low	Low	High
Sinhala Support	No	Pattern-based	Limited	Semantic
Dynamic Updates	Manual	Manual	Manual	Easy
Integration with Plant ID	No	No	No	Planned
Answer Grounding	-	Patterns	Patterns	Retrieved Sources
RAG-based Retrieval	No	No	No	Yes

2 OBJECTIVE

2.1 Main Objective

Sri Lanka is also losing its vast knowledge of medicinal plants. The common names of the medicinal plants in the local languages together with their parts used, their mode of application and the diseases they cure are being recorded either orally in the community or in ancient literature. Unfortunately, with modernization of lifestyle and the death of the elderly population this knowledge is being lost. From available information about 30% of the population is able to identify commonly available medicinal plants around them and that the indigenous knowledge is concentrated in the elderly who are above the age of 55.

Most of the Medicinal plants available around us have no proper information about them and hence this becomes a great challenge for all of us to know all details. Generally, when we see Medicinal plants in our home garden, in the parks, during walks, there are so many questions on our minds like; What is this plant? What is the Sinhala name of this plant? Do we have any medicine from this plant? What disease will be cured using the plant? How should the plant be taken as a medicine? Which part of the plant we need to take as a medicine, leaves, roots, bark etc. All these questions arise on minds at once, but there is no way of getting all the answers at the same time, today.

Printed books are heavy. They represent only a few authoritative but expensive, rare and rather hard to search texts that are unavailable to most of the people. Online information is spread across a wide range of web sites, discussion forums and research papers. As a result, online users have to do the hard work of searching for information, evaluating sources and connecting the dots to form an opinion. Present digital systems either demand highly technical skills (ontologies with SPARQL [1]) or are only capable of a limited amount of knowledge (pattern-matching chatbots [2], [3]). That was also a major obstacle because a lot of information is in English, and the local population here largely communicates in Sinhala.

More specifically, this research seeks to answer:

- Can Retrieval-Augmented Generation (RAG) be used to give the correct, relevant and well-informed answer for any user question in relation to the medicinal plant knowledge available in Sri Lanka, based on references [4] and [6]?

- Can OpenAI's embedding models provide semantic search over English and Sinhala queries without requiring a separate translation pipeline?[5], [6].
- How can we design the system to link to plant identification and have some meaningful feedback from the system regarding the plant?

Main Objective of this Research is Develop a Herbal Knowledge Intelligence System which is a conversational AI assistant application using RAG with OpenAI embeddings and GPT models to fetch correct contextual information about the Sri Lankan medicinal plants in both English and Sinhala languages.

2.2 Specific Objectives

1. Build a Curated Knowledge Base

We want to collect, crawl, index and save the information about 10-20 of the most common medicinal plants in Sri Lanka from the web. We will gather information about the botany and taxonomy of plants including their names in different scripts (English, Sinhala), some descriptions and data about their use as medicines, including their preparation, the parts of the plant used etc. and also the diseases they are used to treat and the applications they are used for. We want to represent this knowledge base in the form of meaningful text snippets that can be embedded and searched.

2. Implement Multilingual Semantic Understanding

An effective query understanding approach is designed based OpenAI's text-embedding-3-small model. This model can help to address the query understanding problem for both English and Sinhala queries by avoiding the traditional pattern-matching approach as well as the translation pipeline-based approaches. The motivation is to understand the actual meaning of diverse ways of asking the same question and bring back useful information whatever the language used.

3. Develop RAG Pipeline with Optimized Retrieval

When we talk about training a model for a Retrieval-Augmented Generation pipeline with tools like LangChain, or other libraries of the sort that have components like embedding the user's

input, and doing a search based on the query in a vector database to retrieve relevant chunks, and using those chunks as context for an LLM to do the actual generation, there are going to be several variables to experiment, like choosing the right chunk size, and the right percentage of overlap between those chunks.

4. Integrate Large Language Model for Response Generation

For Large Language Model search output in the context results, the intended use case is to generate human-sounding responses or answers, leveraging the context found within the search results. For this we will use OpenAI's GPT-3.5-turbo accessed via API. The LLM should ideally respond to the search in the language that the search was performed, and ideally indicate when no results are found or that insufficient information is present to attempt to answer the search query.

5. Design Integration with Plant Identification Component

Implement a simple API that will enable the plant names received from the plant recognition component to be sent to the Herbal Knowledge Intelligence System. So that for each plant identified, it provides relevant answers to the user's subsequent questions about the plant, without the need to enter the name of the plant again.

3 METHODOLOGY

Technologies In this phase, the technologies used in the Herbal Knowledge Intelligence System will be detailed. This system will apply a set of technologies for the search, language analysis and user interface services that will enable the implementation of the conversational system that has been described. Hence, the methodologies employed for this application will integrate artificial intelligence technologies and mobile applications development technologies.

Technologies:

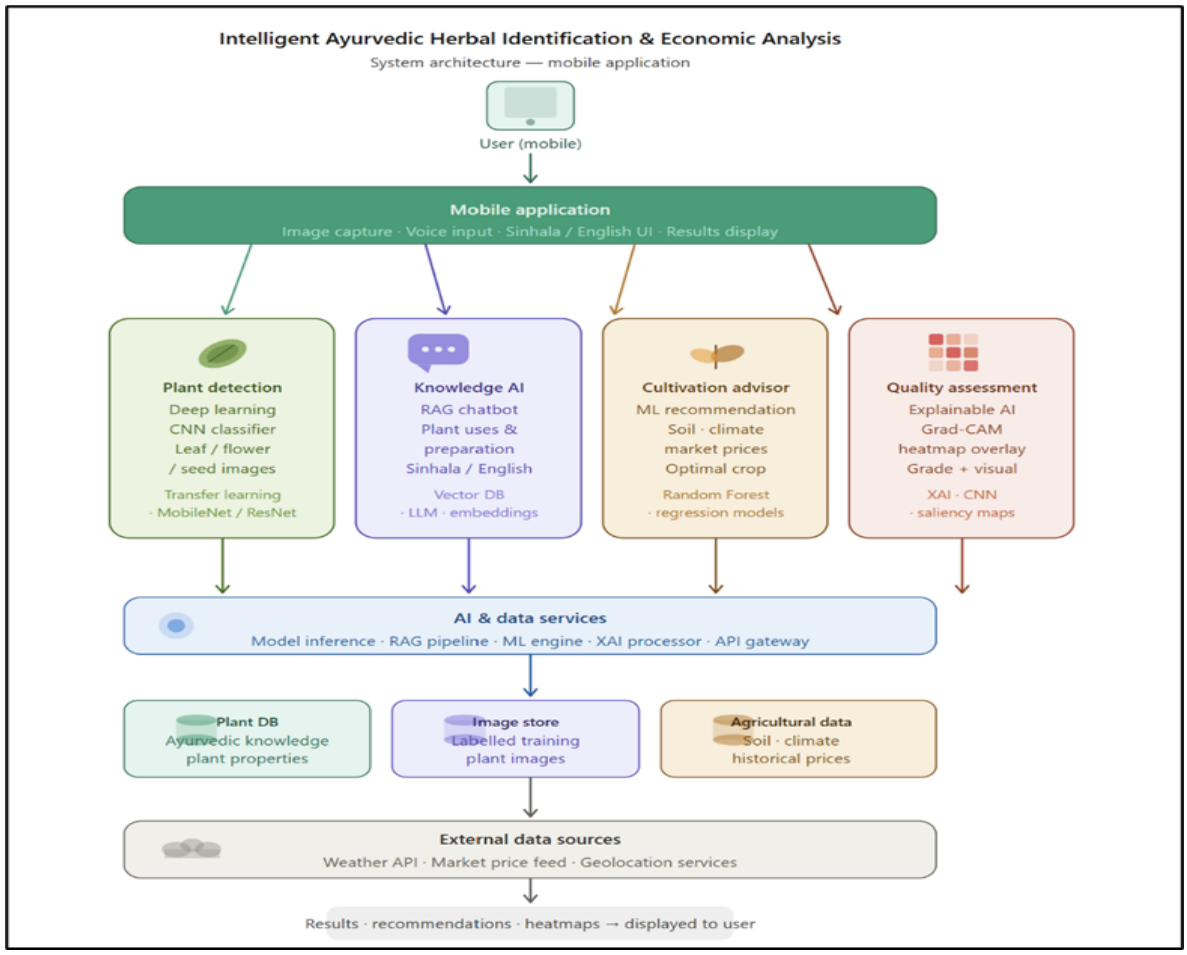
- Knowledge Base Construction: For each plant species, the relevant information will be presented in the following sections: Scientific name, Family, Name in Sinhala, Description, Medicinal applications, Method of preparation, Plant parts used, Treatment of diseases, Applications, Mixtures. After presenting the information for each plant species, the individual sections are split into sub-sequences of length ranging from 256 to 512 tokens using the LangChain.
- RAG Framework: We will be using LangChain to run the RAG pipeline that includes loading and processing the documents (e.g. splitting them into relevant sections, tokenizing, creating embeddings, building the index for retrieval, and sending the final query to the LLM.
- Embedding Model: The system will use OpenAI's text-embedding-3-small mode; for understanding both Sinhala and English queries.
- Vector Database: ChromaDB, an open-source, lightweight, and Pythonic vector store, ideal for prototyping and fast similarity search.
- Large Language Model: The system will use OpenAI's GPT-3.5 access via API for response generation.
- Backend Framework: Flask (Python) for creating a lightweight and flexible API.
- Frontend framework: React Native for building a cross-platform mobile application with a native look and feel.

Validation Strategy:

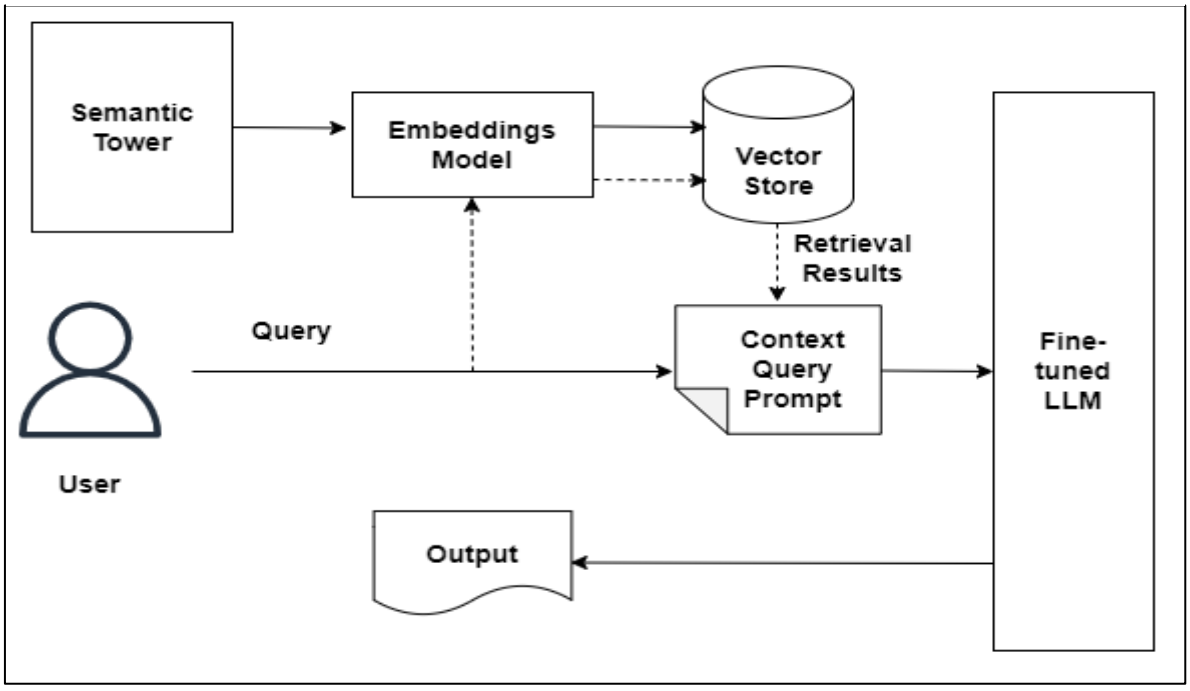
The system's performance will be evaluated using a mixed-methods approach:

- User Experience End users (students, members of the public) will take part in usability testing in order to gain qualitative feedback on the usability of the application, as well as its language support and overall user satisfaction.
- Response Quality: The set of sample questions in both English and Sinhala will be used to get the responses. Those responses will be validated by the domain experts (Ayurvedic Doctors).

4 HIGH-LEVEL SYSTEM ARCHITECTURE



4-1 System Overview Diagram



4-2 Component Overview Diagram

User Query Flow:

- The user asks a question in the application on their mobile device (Example question: “What is Komarika used for? “).
- Mobile app sends a POST request to the /ask endpoint including the query and optional context of the plant to be queried.
- Backend detects language (English/Sinhala) and normalizes text.
- Query is embedded using the same model used for indexing.
- Vector DB performs similarity search, returning top 5 most relevant chunks.
- Retrieved chunks are assembled into a prompt with the user's question.
- LLM generates answer based solely on provided context.
- Response is formatted with answer and sources, returned to mobile app.

5 USER REQUIREMENTS

5.1 Stakeholder Identification

- General Public- The general public seeks information about herbal remedies and traditional treatments. The primary users of the chatbot require straightforward answers which must be both correct and easy to understand.
- Ayurvedic Practitioners and Doctors- Medical professionals who practice traditional Sri Lankan medicine. The system serves as a quick reference tool which enables users to access information about plant properties and preparation methods and medicinal applications.
- Researchers- Academics and scientists who conduct research on Sri Lankan medicinal plants
- Students- The system serves as an educational resource for learners who study Ayurveda or traditional medicine by helping them understand plant properties and medicinal applications.

5.2 Functional Requirements

- FR-01 Natural Language Understanding- Understand user questions in English and Sinhala (text).
- FR-02 Plant Information Retrieval- Provide information about plant name, family, local names.
- FR-03 Medicinal Uses- List diseases treated by each plant.
- FR-04 Preparation Methods Explain- how to prepare plant parts.
- FR-05 provides detailed specification for the various plant parts required for different testing conditions.
- FR-06 Application Categories- Explain how to apply.
- FR-07 Integration with Plant ID- Accept plant name from identification module.
- FR-08 Context Maintenance- Remember conversation context for follow-up questions.

5.3 Non - Functional Requirements

- Performance and Efficiency:

  The system must respond to user's questions in a reasonable time.

  The system shall support up to 10 concurrent users during testing.

  The system shall return relevant chunks for at least 70% of test queries.
- Reliability and Availability:

  System must be available and failure proof to support high concurrency.
- Usability and Accessibility:

  The mobile application must be characterized by a user-friendly and simple interface that is applicable for all age groups.
- Security and Privacy:

  No personal user data shall be stored or shared.

  All API communications shall use HTTPS in production deployment.
- Accuracy:

  All output must be verified to match the true facts as recorded in the knowledge base, with no intentional or unintentional hallucination.

6 COMMERCIALIZATION PLAN

The main target audience for this site would be the Sri Lankan populace in general, specifically adults 25 years and above who are looking for alternatives to modern medication in the form of alternative medicine. There is a secondary target audience which includes tourists, both local and international. The RAG as a technology provides Competitive Advantage as it is able to integrate deep domain knowledge with a flexible, Natural Language based User Interface in both English and Sinhala, whereas static databases and simple chatbots do not provide such flexibility and ease of access to information.

Revenue Model:

Table 6-1 Revenue Table

Model	**Description**	**Estimated Revenue**
Freemium App	Basic info free: premium features (LKR 500/year)	LKR 50,000/year (100 users)
Institutional Licensing	Hospitals/ Schools (LKR 50,000/Year)	LKR 500,000/Year (10 institutions)
Data Insights	Anonymized usage data for research	LKR 100,000/Year
Total Potential Annual Revenue		**LKR 6M – 6.5M**

Cost Estimation: Backend API server/hosting costs (if not local) API costs (in the event of moving to a more powerful cloud LLM) Marketing costs First development costs are based on the cost of human labor.

Table 6-2 Cost Estimation Table

Cost Category	**Initial Development**	**Annual Operating**
Cloud Hosting	LKR 20,000	LKR 60,000
API Services	LKR 10,000	LKR 40,000
Maintenance	-	LKR 100,000
Total	**LKR 30,000**	**LKR 200,00**

7 BUDGET AND JUSTIFICATION

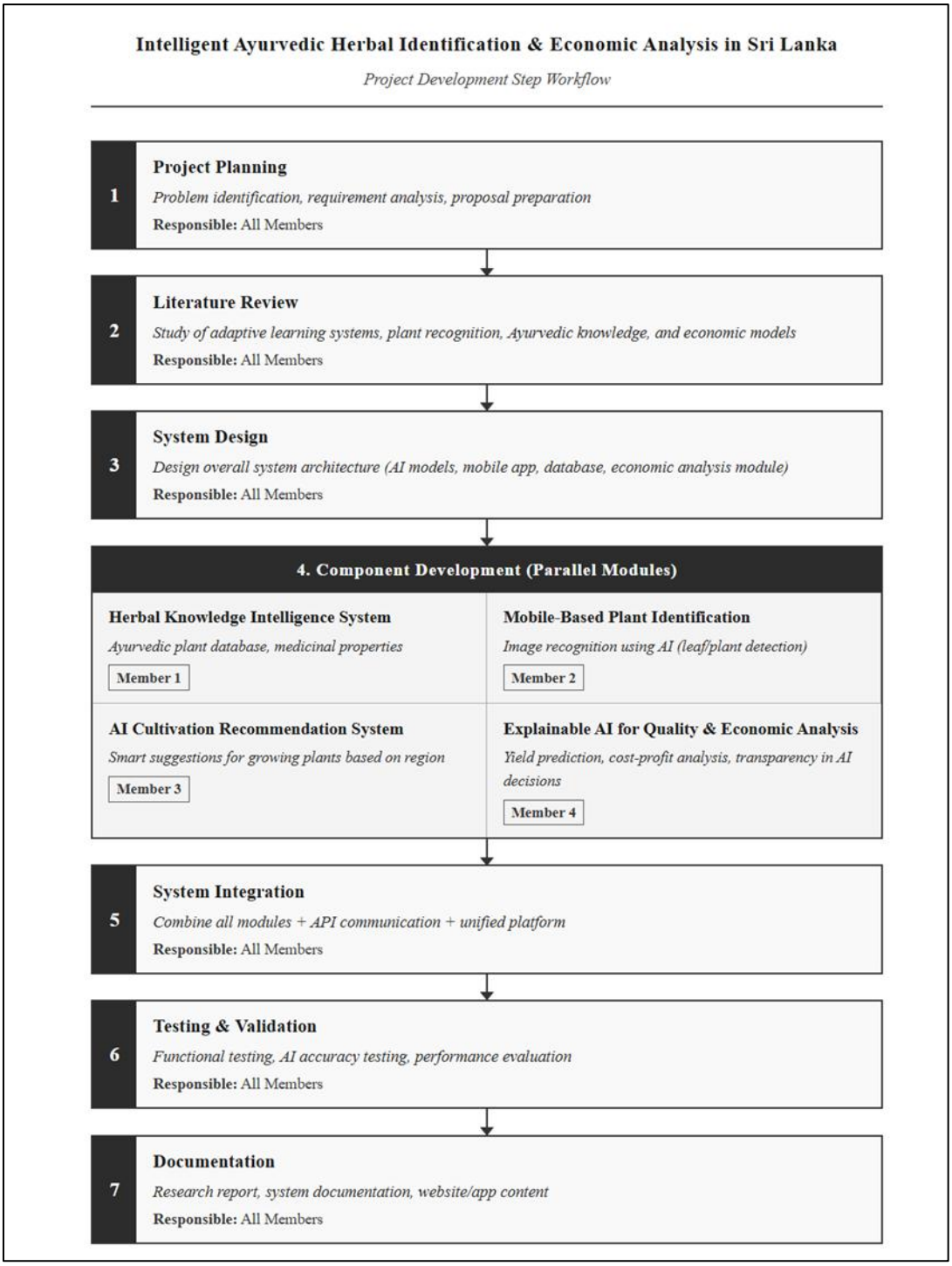
Table 7-1 Estimated Budget Table

Description	Total Amount (Rs.)
Software and Tools (API Services/ IDE licenses, development tools)	10,000.00
Cloud Computing and Hosting	12,000.00
Paper Publishing	15,000.00
Transportation and Communication	5000.00
Printing and Documentation	3000.00
Total	**48,000.00**

Budget Justification:

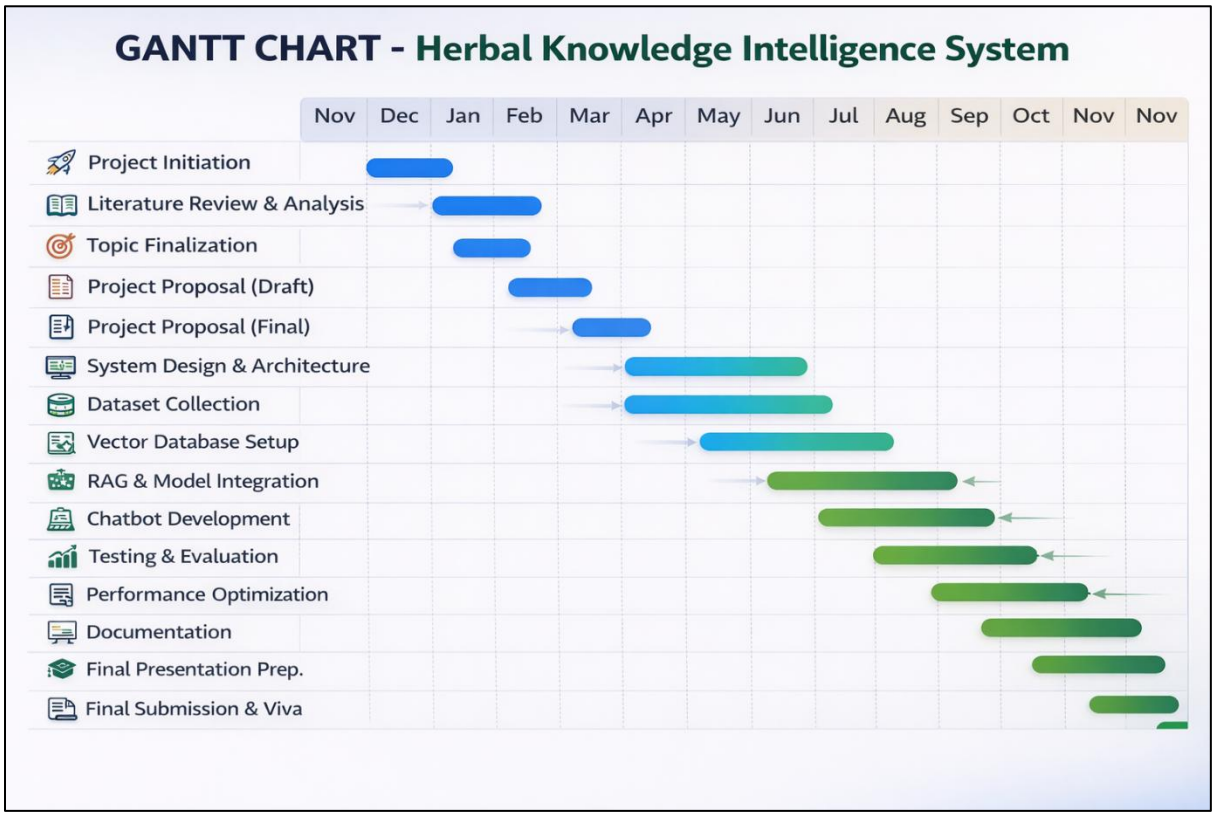
- **Software and Tools (LKR 10,000):** Covers any premium development tools, API access and potential costs for mobile app distribution.
- **Cloud Computing and Hosting (LKR 12,000):** Cost for cloud GPU usage if local resources are insufficient for model inference, or for hosting the backend API during testing.
- **Transportation and Communication (LKR 5000):** Travel costs for consultations with supervisor meetings and ay data collection visits.
- **Printing and Documentation (LKR 3000):** Printing of reports, presentation materials and other documentation needs.
- **Paper Publishing (LKR 15,000):** If the research findings are accepts for publication in a conference, publication fees may apply.

8 WORK BREAKDOWN STRUCTURE (WBS)



8-1 Work Breakdown Structure

9 GANTT CHART



9-1 Gantt Chart

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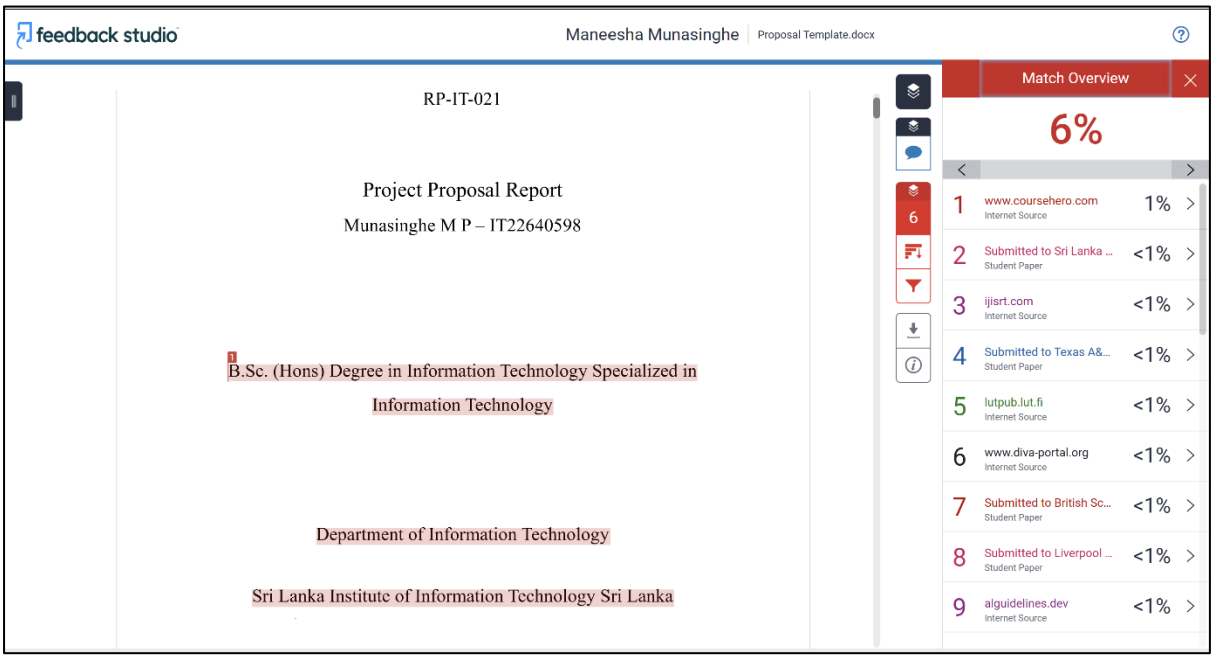
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11 APPENDICES



Appendix 1- Plagiarism Report